	ITER, SIKSHA 'O' ANUSANDHAN **(Deemed to be University** Interdependence and Gains from Trade; Measuring Cost of Living**)**		**Assignment I**
Branch	**ALL**	Programme	B. Tech.
Course Name	Supply Chain Management Strategy	Semester	**8th**
Course Code	OPL4004	Academic Year	2025-26
Assignment: 1		**Topic:** Introduction to Supply Chain Management, and Supply Chain Strategy	
Learning Level (LL)	**L1**: Remembering	**L3:** Applying	**L5**: Evaluating
	L2: Understanding	**L4**: Analysing	**L6**: Creating

Q's	Questions	COs	LL
1	"SCM encompasses all activities involved in the transformation of goods from the raw material stage to the final stage, and ultimately reaching the end customer." Analyse this statement with the help of a suitable diagram.	**CO1**	**L3**
2	"Bubble" is a firm engaged in the manufacturing of shirt. To produce 100 shirts in a day, it purchases 300 meters of woven fabrics from 'ZItalia', 1500 buttons from 'MButtons', 100 collars from 'JCollars', 100 bundles of thread from 'SThread', 200 labels from 'VLabel', and 100 packaging boxes from 'Bbox'. Using these inputs 'Bubble' produces 100 shirts. After production, the shirts have been transported to its Central Warehouse and then to its 10 Retail Outlets for its sale to final consumers. Draw a supply chain diagram showing the flow of Bubble shirts from the supplier to the consumer, indicating upstream and downstream activities. Treat 'Bubble' as the focal firm.	**CO1**	**L3**
3	You are a focal firm engaged in the manufacturing of study tables. To produce 100 study tables in the month of December 2025, you purchase wood and ply from supplier-1, steel from supplier-2, wood adhesive from supplier-3, and screw, nuts and bolts from supplier-4. After the completion of the production process, you supply the study tables to the consumer through wholesaler 'W1' and the retailer 'R1'. Draw a supply chain diagram showing the flow of study tables from the supplier to the consumer, indicating upstream and downstream activities.	**CO1**	**L3**
4	"Is a commodity position always bad?". Analyse using the competitive advantage matrix.	**CO2**	**L3**
5	***"COVID-19*** pandemic witnessed disruptions in the supply chain management of all most all commodities in India". Analyse which of the two effects, viz., Ratchet or Ripple better explains this disturbance in SCM.	**CO2**	**L3**
6	A warehouse introduces barcode scanning to reduce manual errors. Analyse the likely impact of introduction of barcode on productivity in the supply chain.	**CO2**	**L3**
7	Logistics refers to "planning, implementing, and controlling the physical flows of materials and finished goods from point of origin to point of use to meet the customer's need at a profit". Analyse this statement in the context of relationship between logistics and supply chain management.	**CO1**	**L3**
8	Delta Pizzaria is a local pizza shop in Bhubaneswar which operates 10 hours a day with the help of 20 workers. Each day, it can be able to produce and sale 500 pizzas. It is	**CO2**	**L3**

	estimated that on an average 2 pizzas has been produced in each labour hour by all the pizza shops in the city. Analyse, whether Delta Pizzaria enjoys competitive advantage in the production of pizza, measured through its labour productivity.		
9	A supply chain facility produces 5,000 units using the following inputs: (a) Labor: 400 hours at ₹200/hour (b) Machine: 300 hours at ₹150/hour (c) Energy cost: ₹20,000. Next month, output increases to 6,000 units with (a) Labor: 420 hours at ₹200/hour (b) Machine: 320 hours at ₹150/hour (c) Energy cost: ₹22,000. Analyze the change in total factor productivity (TFP).	**CO2**	**L4**
10	A supply chain facility increases output from 1500 units to 1800 units, while labor hours increase from 300 to 320 hours. Analyze the change in labour productivity.	**CO2**	**L4**

Note:

1. Assignment-I carries a weightage of 4 marks.
2. The course outcomes 1 and 2 were covered.

Course Outcomes	CO1	Students will be able to explain the concept, flows, characteristics, strategic importance, decisions, and the boundary-spanning role of supply chain management.
	CO2	Students will be able to analyze supply chain strategies, network design decisions, and the alignment between business strategy and supply chain strategy in explaining competitive advantage.